

WASP-140b

R-2249

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-140_201216 · created 2026-08-15T23:45:37+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459199.7629 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.065 +/- 0.048
Transit depth (Rp/Rs)^2	0.43 +/- 0.62 [%]
Orbital Inclination (inc)	83.83 +/- 2.6
Airmass coefficient 1 (a1)	0.98 +/- 0.007
Airmass coefficient 2 (a2)	0.011 +/- 0.011
Scatter in the residuals of the lightcurve fit is	0.71 %
Best Comparison Star	#1 - [237, 292]
Optimal Aperture	5.33
Optimal Annulus	14.29
Transit Duration (day)	0.04 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.065 ± 0.048 vs 0.1432 (deviation 1.63σ)
Duration fit vs published	0.04 d vs 0.0604 d (deviation 0.76σ)
Mid-time O–C	-1.5 min
Depth SNR	1.4
Residual scatter	0.71 %

Light curve

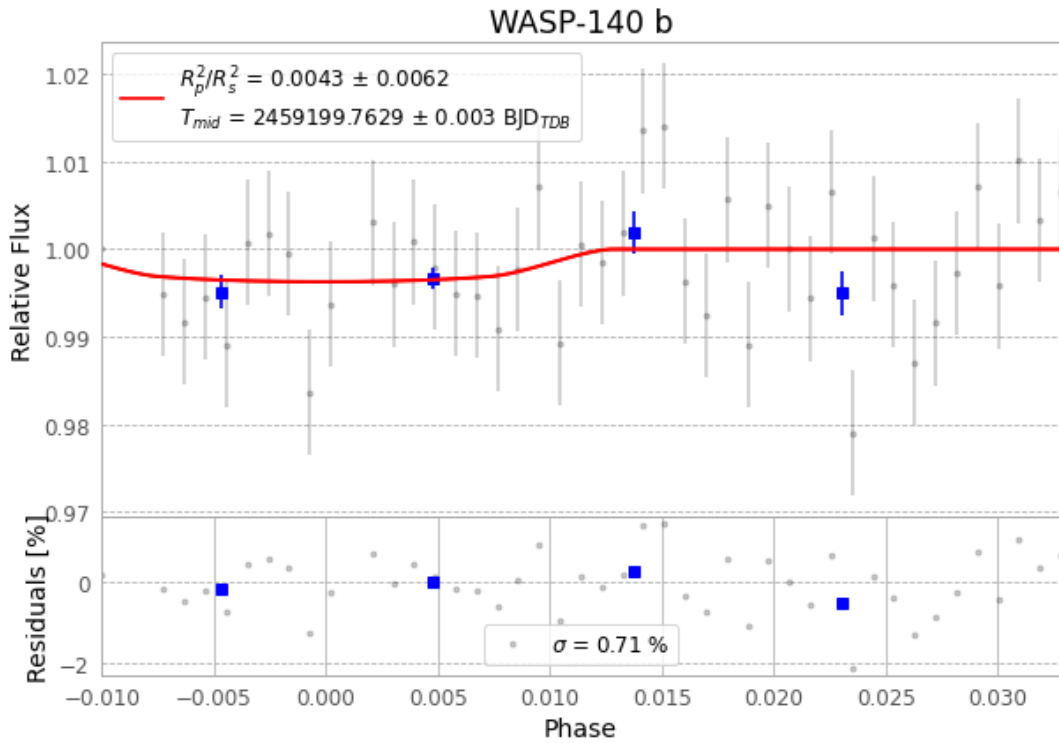


Figure 1 — FinalLightCurve_WASP-140 b_2020-12-15.png (SHA-256 6a4ab5cd976788e7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [255, 260], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a0619c0be55579d5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

45 FITS frames (30 MB), WASP-140201216053912.FITS → WASP-140201216075712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-140-201216.log (a33eb46fa5d9...), FinalLightCurve_WASP-140 b_2020-12-15.png (6a4ab5cd9767...), FinalLightCurve_WASP-140 b_2020-12-15.csv (4cc56536787d...)

Compute resources

Wall time 87.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 23:45Z.